NETWORKING



The Disappearing DSP

Operating Linear and Co-Packaged Optics at Hyperscale

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The operating question is not DSP or no DSP

Preserve the operational contract — even if the DSP disappears.

Physics stays

Noise, loss, reflections, thermal drift and marginal lanes still exist.

The boundary moves

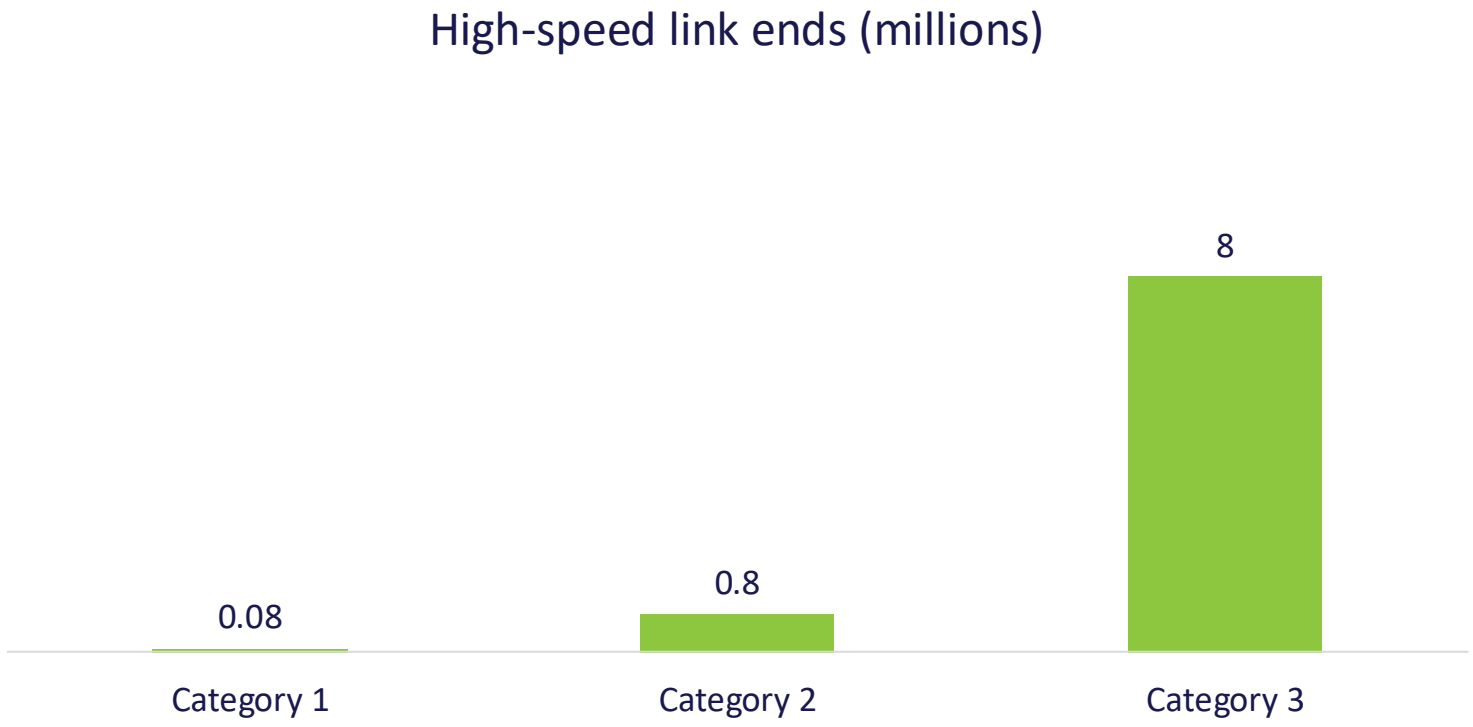
LPO distributes the analog burden. CPO integrates it into a larger fault domain.

Operations must follow

Telemetry, identity and remediation must cross the link — and survive resets.

Hyperscale operations must maintain observability

Radix turns rare failures into routine operations



Illustrative scale
8 high-speed links / host

10K → 80K link ends
100K → 800K
1M → 8M

Fleet rate ≈ population × per-link event rate

Illustrative topology — the multiplication is the point.

The DSP is more than a signal processor

RETIMER

Recovers timing and reshapes the electrical signal.

EQUALIZER

Absorbs channel loss and compensates for frequency dependent effects.

FEC / BER

Counts impairment before it becomes a link-down event.

BLACK BOX

Keeps diagnostics, state and firmware close to the physics.

FAULT LINE

Creates a replaceable, attributable operational boundary.

Remove the silicon if the architecture demands it. Do not accidentally remove these five functions from the operating model.

Removing the DSP does not remove the physics

LPO — distributed analog path

Shorter, lower-power module path
Host SerDes carries more equalization burden
Electrical channel + optics become one coupled margin budget
Fault evidence is split across ASIC, board and module

CPO — integrated optical engine

Very short electrical reach
Optics move into the switch/package fault domain
Lasers, fiber attach and connectors reshape serviceability
A single physical event can have a wider blast radius

Common requirement: correlate electrical + optical + remote evidence

The field surprise: less DSP did not mean less reliable

Like-for-like populations

Sanitized operator data did not show LPO–LPO links as less stable than comparable retimed populations.

Directionality mattered

Mixed-vendor behavior was asymmetric: transmitter/receiver pairing could dominate the outcome.

Attribution got harder

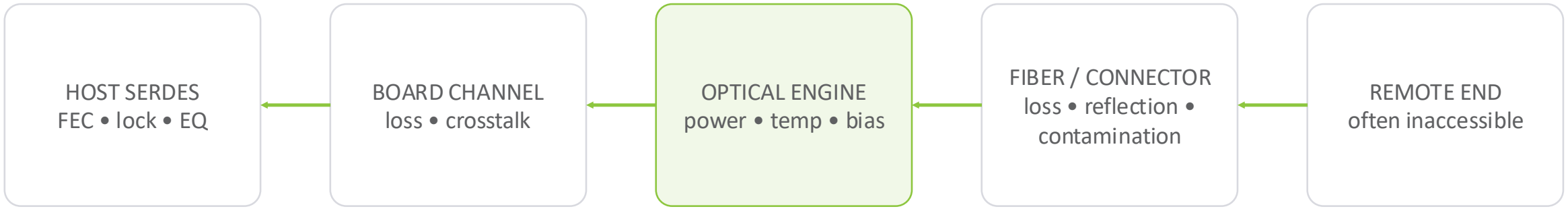
A down transition proves impact. It does not identify host, electrical path, media path or remote endpoint.

Reliability can improve while explainability regresses.

The edge of observability is moving away from the physics

Owned endpoint: deep counters

Far endpoint: inference unless telemetry crosses the link



The module DSP used to collapse this chain into one operational box. DSP-less designs must rebuild the correlation explicitly.

Host links are where blindness hurts most

OWNED SWITCH

Full NOS access
ASIC + module telemetry
Fast local actions

host-facing optical link

BARE-METAL HOST

Limited or no agent access
Remote counters may be absent
Repair starts from the switch side

LOCAL / HOST
SerDes • board

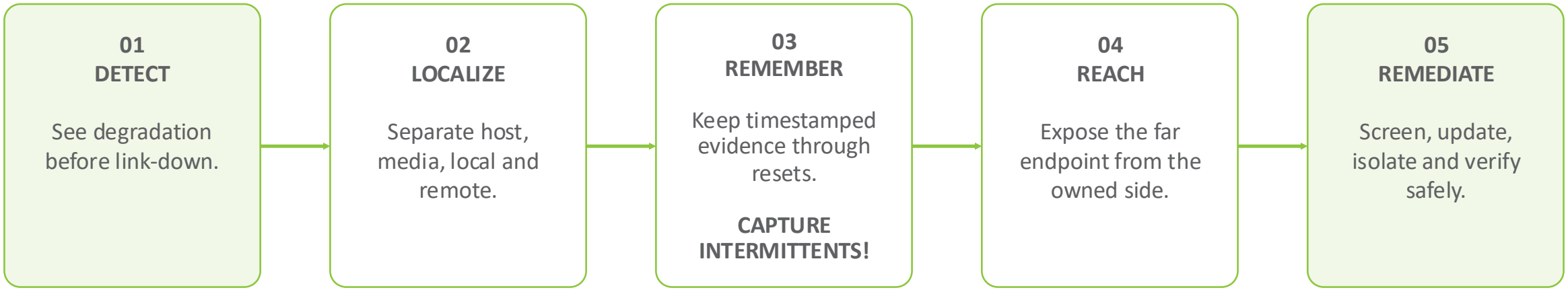
LOCAL / MEDIA
TX/RX • connector

REMOTE / HOST
NIC SerDes • board

REMOTE / MEDIA
remote optic • fiber

You cannot localize four failure domains with one endpoint's counters.

Preserve the operational contract, not the component



This contract should be testable in qualification and portable across LPO, CPO and retimed optics.

Link-aware optics closes the control loop



Demonstrated capability set

- local / remote × host / media localization
- remote module telemetry and identity
- timestamped event logs
- link screening and automatic shutdown
- remote and local firmware upgrade

The POC used DSP-to-DSP messaging. The operational pattern can survive the DSP.

OCP: define the observability contract

OPTICS RELIABILITY WORKSTREAM — TECHNICAL SCOPE

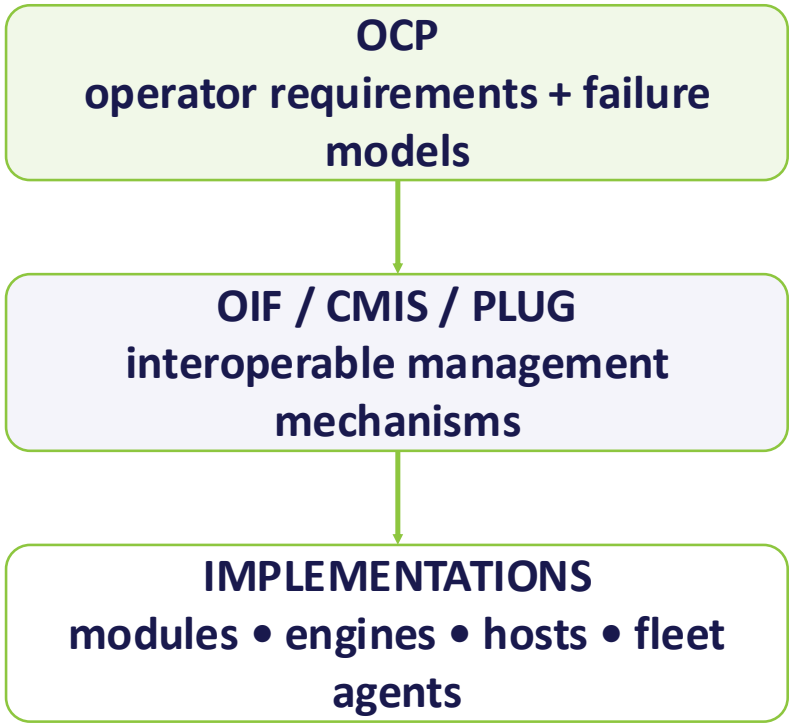
Predict and prevent link flaps

Timestamp event logs using standardized module time

Reach the far-end module from the owned endpoint

Carry module-to-module messages in-band

Define portable commands — avoid vendor custom-ID collisions



Run requirements and protocol work in parallel — not serial.

Telemetry itself must survive hyperscale

POLL LESS

Use capability discovery and adaptive cadence.

REMEMBER LOCALLY

Keep bounded event rings near the physics.

ESCALATE ON CHANGE

Push transitions and threshold crossings.

QUERY BY INTENT

Fetch deep detail only after classification.

FAST PATH — protect the link

Local thresholding
Event trigger
Safe autonomous action

SLOW PATH — learn from the fleet

Correlate endpoints
Compare populations
Tune thresholds and verify repairs

EBO mitigates mechanics — not the observability

WHAT EXPANDED BEAM CAN MITIGATE

Larger effective beam at the interface
Lower sensitivity to a single dust particle
Lower mating force in high-fiber-count interfaces
A path to more serviceable fiber attach

WHAT STILL REQUIRES QUALIFICATION

Insertion-loss and alignment budget
Connector-cycle durability and wear
Cleaning/inspection practice
Multi-vendor interoperability
Blast radius of a shared connector

Mechanical resilience reduces incidence. Telemetry reduces time-to-repair.

Help define the observability contract

Join the OCP Networking / Optics Reliability workstream

Review the workstream wiki and Optics Telemetry draft

Contribute operator failure modes, counters and interoperability cases

Align OCP requirements with OIF CMIS / PLUG mechanisms

Bring LPO, CPO, host and fleet-tool perspectives before interfaces harden

Wiki: opencompute.org/w/index.php?title=Networking/OpticsReliability

Draft: opencompute.org/documents/2025-ocp-software-specification-optics-telemetry-rev0-9-pdf

The DSP may disappear.

Its operational responsibilities cannot.

